One engine for LatentMAS

vLLM fork · latent steps + Judger decode · Qwen3 · one H200

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The gap

- Paper numbers use the **Hugging Face** LatentMAS path.
- Authors' "vLLM support" still runs latent agents on a second HF copy, then hands off via `prompt_embeds` and a re-prefill.
- That handoff drops the in-engine KV and blocks the Judger on a CPU/GPU transfer.

Contribution: a vLLM 0.11.1 fork where Planner / Critic / Refiner latent steps and Judger decode share one request and one KV.

What the fork does

- Flag a request with a phase schedule: latent $\rightarrow$ role tokens $\rightarrow \ldots \rightarrow$ decode.
- Latent step: next input is `realign(last_hidden)` written into `inputs_embeds`, not a sampled token embedding.
- Role prompts append as ordinary tokens on the *same* request.
- Latent embeddings are stored so preemption / recompute replays them.
- Start with `enforce_eager`, async scheduling off; stock or tail scheduler via `scheduler_cls`.

Correctness vs Hugging Face

- Prompt token ids match; forced preemption replays stored latent embeds.
- First latent of each agent: cosine ≈ 0.999 (role-prompt splice OK).
- All $3K$ steps: recurrent feedback amplifies attention-kernel noise.

14B AIME control (min cosine over 30 steps, item 0): HF sdpa vs eager 0.81; HF vs fork 0.82. Fork drift is inside the HF kernel-drift envelope — not a separate realign/RoPE bug.

What we measure next

- Primary: unified fork vs HF-only LatentMAS (paper setup).
- Also: repaired hybrid, single-model vLLM, text-MAS vLLM.
- AIME 2024+2025, MATH-500 (n=200), GSM8K (n=200), Qwen3-14B.
- Concurrency 1/8/16/32/60 closed loop for vLLM arms.
- Throughput, p50/p95 latency, accuracy with paired bootstrap CIs, memory, preemptions.
- Greedy and sampled kept separate; single-model always reported.

GSM8K: same accuracy, lower latency

n=50, greedy, Qwen3-14B, C=1, K=10, max new 2048.

arm	accuracy	p50 latency	tok/s
HF-only LatentMAS (paper)	0.98	13.3 s	24.3
unified vLLM fork	**0.98**	**5.8 s**	**86.4**
single-model vLLM	0.98	6.5 s	68.3

Paired Δ accuracy 0.0 pp, CI $[-6, +6]$. HF/fork latency ratio $\approx 2.2\times$. At C=32 the fork reaches 776 tok/s (wave-batched).

AIME: faster, but not more accurate

AIME 2024+2025, n=60, greedy, max new 8192.

arm	accuracy	p50	tok/s
HF-only	**0.53** (32/60)	175 s	24.6
unified fork	0.38 (23/60)	94 s	86.8

Fork wall 1.99× lower, but paired accuracy $\Delta = -15$ pp, CI $[-27, -5]$. **34/60** fork items hit the 8192 Judger length cap (HF: 0/60, max $\approx$ 6474). Do not claim an AIME accuracy win.

Concurrency (wave-batched)

Threaded concurrent `generate` segfaults (TMA). One wave per `generate` is stable.

arm	C	tok/s	p50 s
fork_stock	1	86	5.8
fork_stock	8	275	2.7
fork_stock	32	776	0.68
fork_stock	60	1146	0.51
fork_tail	32	769	0.69

Tail vs stock within noise when the KV pool is not overflowing. Accuracy stays ≈ 0.98 on GSM8K.

Takeaways

- One engine removes the hybrid handoff; GSM8K matches HF accuracy at $\sim 2\times$ lower p50.
- Single-model vLLM is reported (0.98 acc, 6.5 s p50, 68 tok/s) — not hidden.
- AIME: speed win, accuracy loss under the same length budget; length caps dominate.
- Per-step latents match HF within normal SDPA/eager kernel drift on 14B.